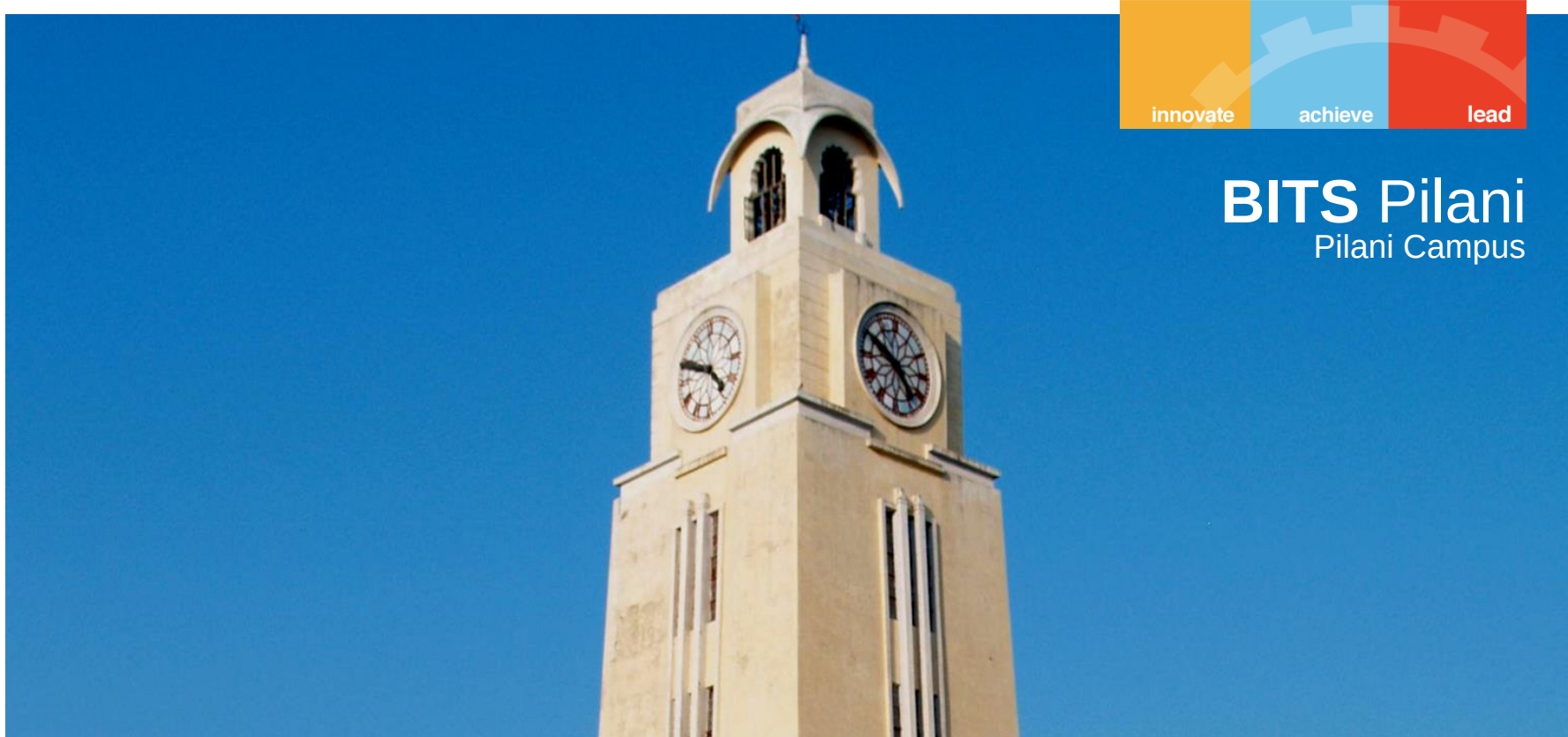




BITS Pilani
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CHEM F111 : General Chemistry

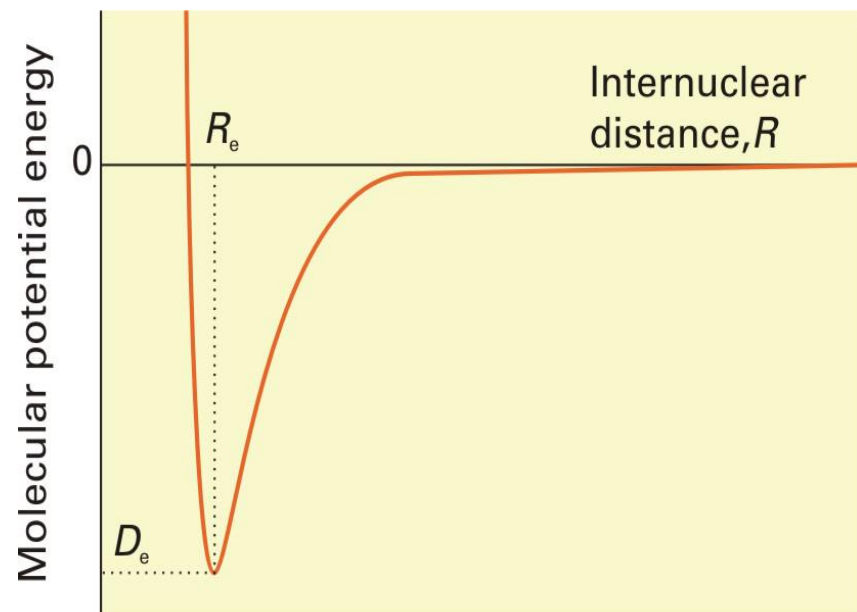
Semester II: AY 2023-24

Lecture- 12, 9th February 2024, Friday

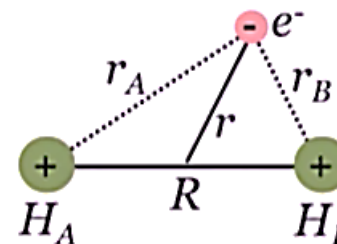
Last Class:

- Some ideas about Valence Bond Theory; Hybridization and Resonance;
- MO Theory; Linear combination of atomic orbitals;
- Energy determines the orbitals being bonding, anti-bonding or non-bonding;
- Wave functions of H_2^+ ; bonding and anti-bonding orbitals; overlap integral;
- σ and σ^* orbitals; numbering scheme; gerade and ungerade symbols for molecules having center of inversion;

Recap: Molecular orbital theory [for diatomic molecules]



The Hamiltonian operator for H_2^+



$$\hat{H}(\text{H}_2^+) = -\frac{\hbar^2}{2m_A} \nabla_A^2 - \frac{\hbar^2}{2m_B} \nabla_B^2 - \frac{\hbar^2}{2m_e} \nabla_e^2 - Q \frac{e^2}{r_A} - Q \frac{e^2}{r_B} + Q \frac{e^2}{R}$$

- The Hamiltonian operator for H_2^+ in Born Oppenheimer (B.O.) approximation:

$$\hat{H} = -\frac{\hbar^2}{2m_e} \nabla^2 - \frac{e^2}{4\pi\epsilon_0 r_A} - \frac{e^2}{4\pi\epsilon_0 r_B} + \frac{e^2}{4\pi\epsilon_0 R}$$

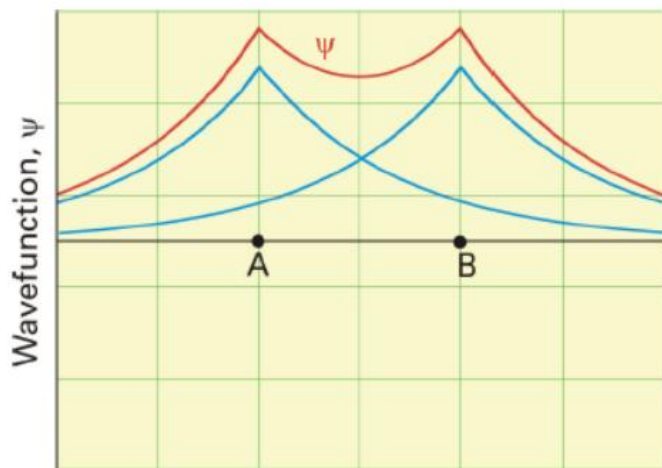
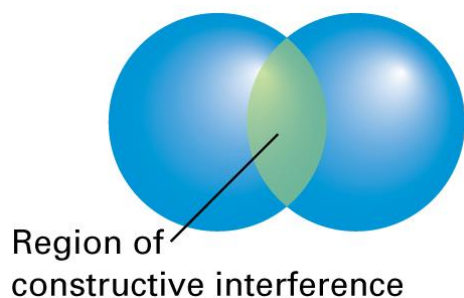
Recap: Molecular orbital theory [for diatomic molecules]



Bonding

$$\Psi_1 = N(\phi_A + \phi_B)$$

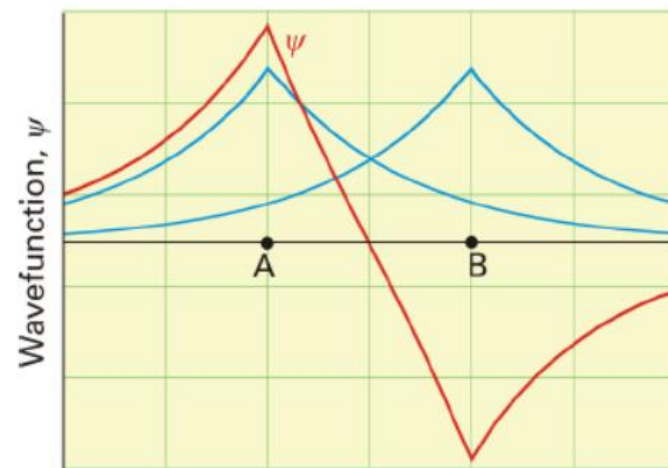
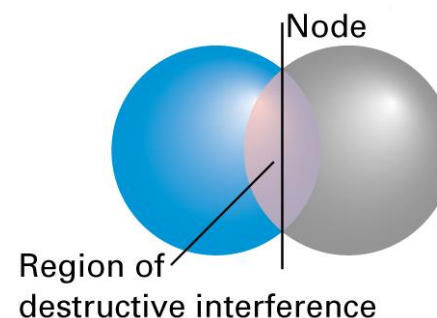
$$\Psi_1^2 = N^2 (\phi_A^2 + \phi_B^2 + 2 \phi_A \phi_B)$$



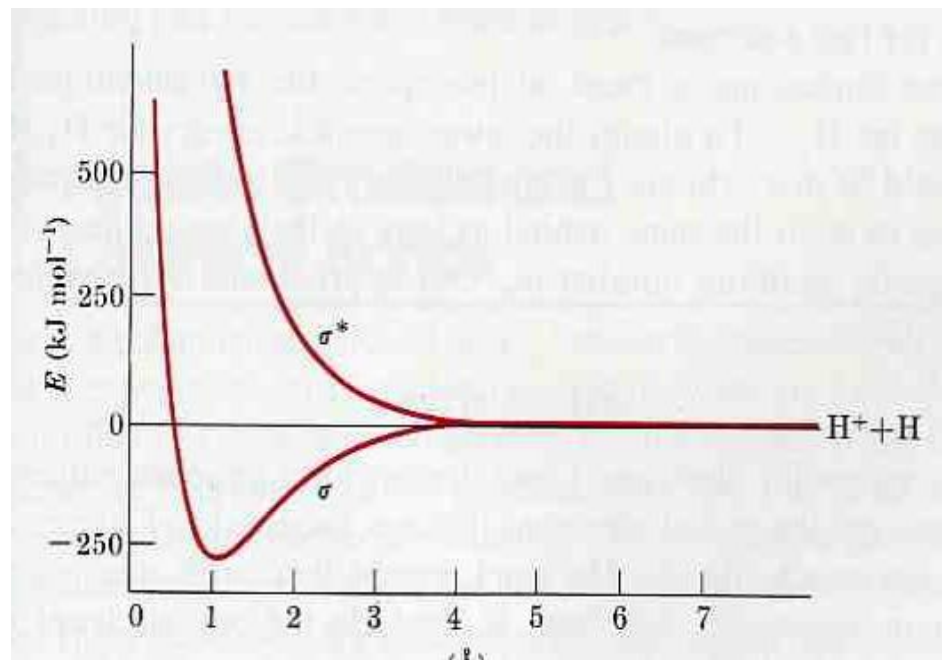
Anti-bonding

$$\Psi_1 = N(\phi_A - \phi_B)$$

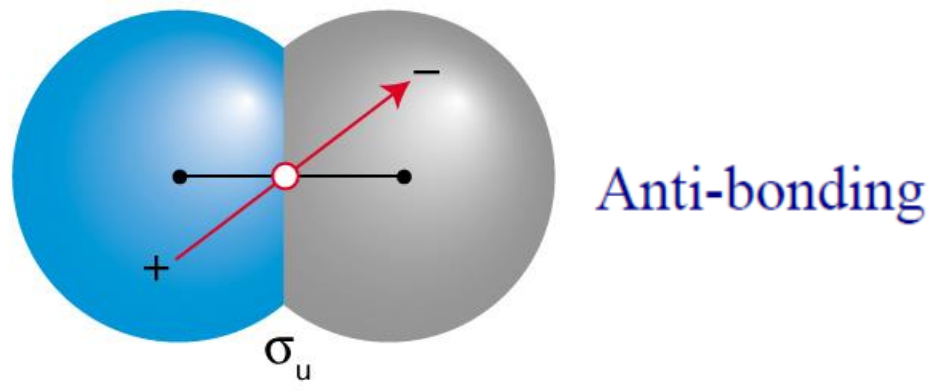
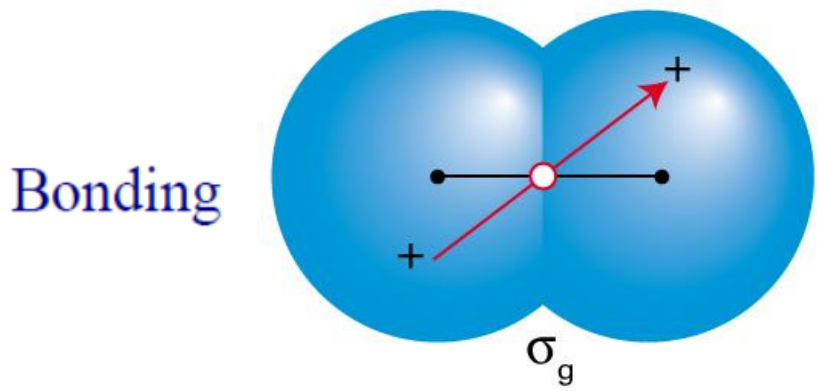
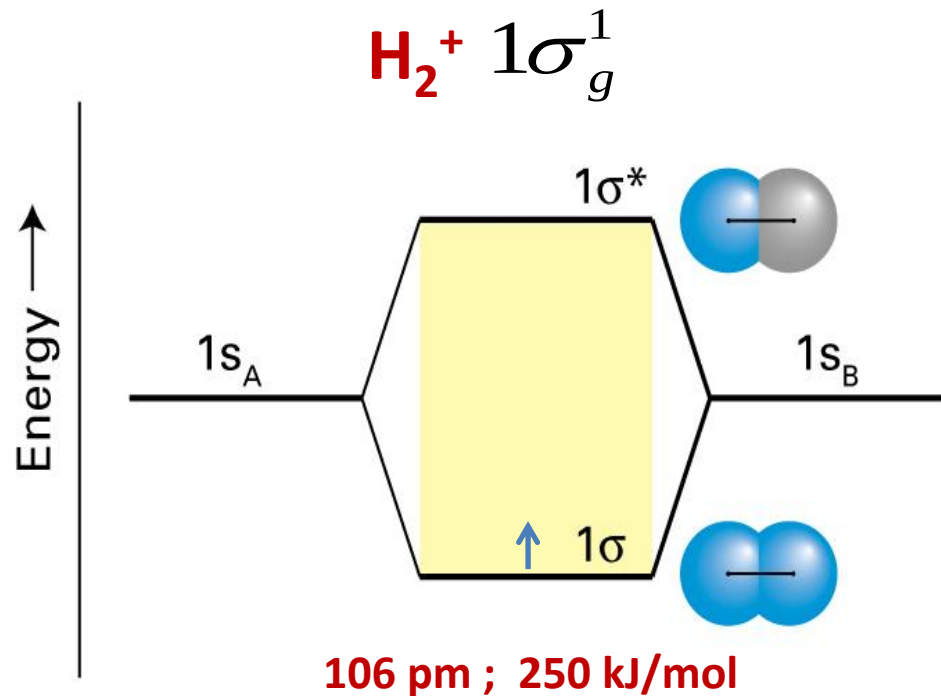
$$\Psi_1^2 = N^2 (\phi_A^2 + \phi_B^2 - 2 \phi_A \phi_B)$$

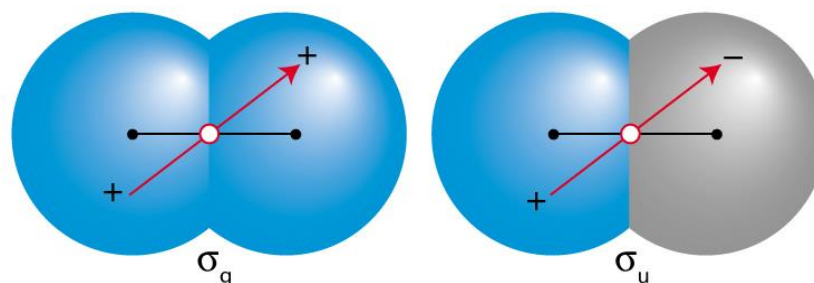


Molecular Orbital Theory [diatomic molecules]

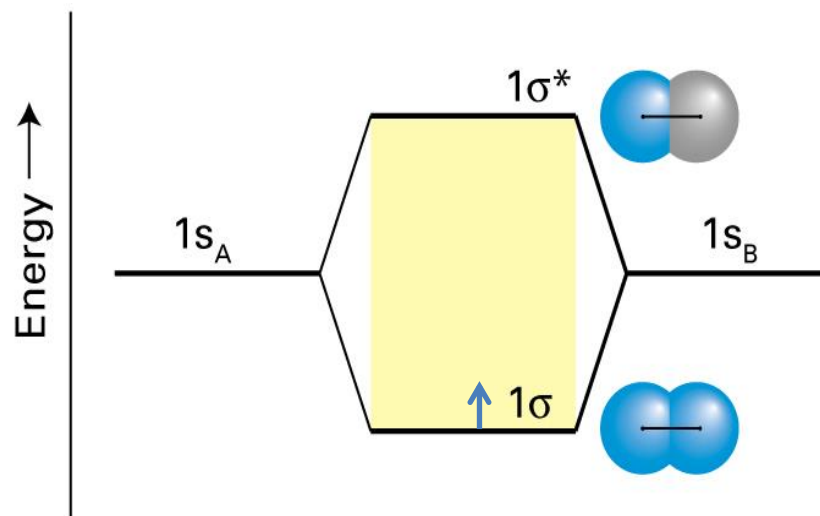
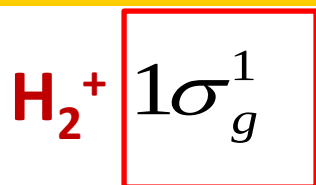


Inter nuclear Separation





- *Gerade (g) and ungerade (u) symbols are based on symmetry w.r.t. Centre of Inversion.*
- *Centre of Inversion: (x,y,z) and $(-x,-y,-z)$ have same properties*
- *These symbols are not applicable for molecules without centre of inversion such as CO, Methane etc.*



MO's characterized by:

- Bonding / Anti-bonding character – lowering or raising the energy
- Component of electronic angular momentum along the axis passing through the nuclei:

0 (for σ), $\hbar$ (for π), $2\hbar$ (for δ) – number of nodal planes containing axis.

- Transformation under inversion – symmetric (g), anti-symmetric (u)

(Hetero-nuclear diatomic molecules lack this symmetry element.)

In the above example of H_2^+ molecule, we obtain only σ_g and σ_u orbitals

H_2^+ Given that, $\Psi_1 = N_+(\psi_A + \psi_B)$

is the bonding orbital of H_2^+ , let us find out the normalization constant N_+

- We know that: $\int \psi^* \psi d\tau = 1$

Substitute the LCAO into the above integral, considering atomic orbitals are individually normalized.

$$\int \psi^* \psi d\tau = N_+^2 \left\{ \overbrace{\int \psi_A^2 d\tau}^1 + \overbrace{\int \psi_B^2 d\tau}^1 + \overbrace{2 \int \psi_A \psi_B d\tau}^S \right\} = 2(1+S)N_+^2$$

So, $N_+ = \frac{1}{\{2(1+S)\}^{1/2}}$ where “S” stands for overlap integral.

Q: Find out the normalization constant of antibonding orbital of H_2^+

Molecular Orbital Theory

H₂⁺ [Derivations not in syllabus] *[Understand only the physical significance of the terms]*

The bonding orbital energy is:
$$E_{1\sigma} = E_{H1s} + \frac{e^2}{4\pi\epsilon_0 R} - \frac{j+k}{1+S}$$

The antibonding orbital energy is:
$$E_{2\sigma} = E_{H1s} + \frac{e^2}{4\pi\epsilon_0 R} - \frac{j-k}{1-S}$$

$j_0 = e^2/4\pi\epsilon_0$

$$S = \left\{ 1 + \frac{R}{a_0} + \frac{1}{3} \left(\frac{R}{a_0} \right)^2 \right\} e^{-R/a_0}$$

$$j = \frac{e^2}{4\pi\epsilon_0 R} \left\{ 1 - \left(1 + \frac{R}{a_0} \right) e^{-2R/a_0} \right\}$$

$$k = \frac{e^2}{4\pi\epsilon_0 a_0} \left(1 + \frac{R}{a_0} \right) e^{-R/a_0}$$

E_{H1s} is the energy of a H1s orbital

j₀/R is the potential energy of repulsion between the two nuclei

j is a measure of the interaction between a nucleus and electron density centered on the other nucleus

k is a measure of the interaction between a nucleus and the excess electron density in the internuclear region arising from overlap

- 1. Linear Combination of “suitable” valence orbitals of atoms; N atomic orbitals should give N Molecular orbitals.*
- 2. Accommodate the electrons without violating Pauli principle.*
- 3. In degenerate orbitals, spin pairing should be done after all such orbitals are singly occupied with parallel spin (Hund’s rule).*

Simple MO picture of H₂ molecule

innovate

achieve

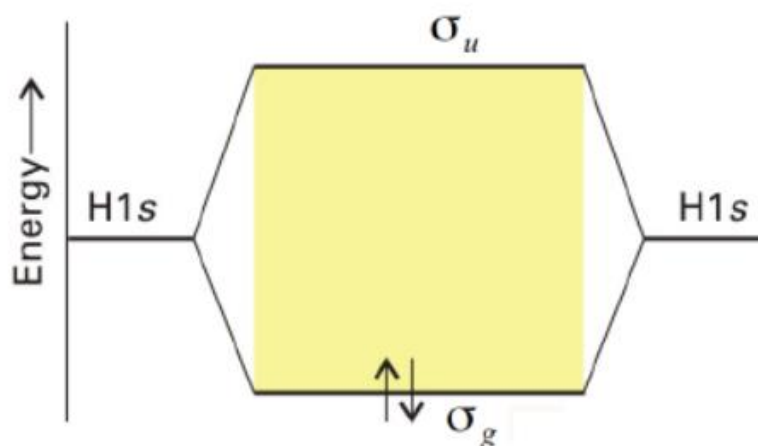
lead

Use the molecular orbitals of H₂⁺, and use the aufbau principle to get the electronic configuration:

- Place both the electrons in the bonding MO, ϕ_1 .
- The electronic configuration would be σ_g^2 .
- The ground state wavefunction would be

$$\psi_{H_2}^{g.s.}(1,2) = \frac{1}{\sqrt{2}} \sigma_g(1) \sigma_g(2) [\alpha(1)\beta(2) - \beta(1)\alpha(2)]$$

This is in fair agreement with experiment



74 pm ; 436 kJ/mol

Simple MO picture of H₂ molecule

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The ground state wavefunction of H₂ molecule is

$$\psi_{H_2}^{g.s.}(1,2) = \frac{1}{\sqrt{2}} \sigma_g(1) \sigma_g(2) [\alpha(1)\beta(2) - \beta(1)\alpha(2)]$$

Now we expand the MOs in terms of linear combination of the atomic orbitals and write the spatial part of the ground state wavefunction and rearrange to get

$$\psi_{H_2}^{g.s.}(1,2)_{\text{spatial}} = \sigma_g(1) \sigma_g(2)$$

$$= N_g^2 (1s_A + 1s_B)(1) (1s_A + 1s_B)(2)$$

$$= N_g^2 [1s_A(1)1s_B(2) + 1s_B(1)1s_A(2)] \quad \text{Covalent interaction}$$

$$+ N_g^2 [1s_A(1)1s_A(2)] \quad \text{Ionic interaction}$$

$$+ N_g^2 [1s_B(1)1s_B(2)] \quad \text{Ionic interaction}$$

Homework: Write the wavefunction for the doubly-excited state of H₂ molecule (electronic configuration: σ_u^2). Using $\sigma_u = N_u(1s_A - 1s_B)$, expand the spatial part, rearrange and see what you get.

Bonding, anti-bonding and non-bonding MO's and bond-order



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Bonding orbitals – atomic orbitals overlap constructively; Probability density increases in the region between the two nuclei; energy lower than that of the constituent atomic orbitals. No nodal plane between the two nuclei (perpendicular to the internuclear axis) e.g. σ_g , π_u , δ_g , ..., orbitals.

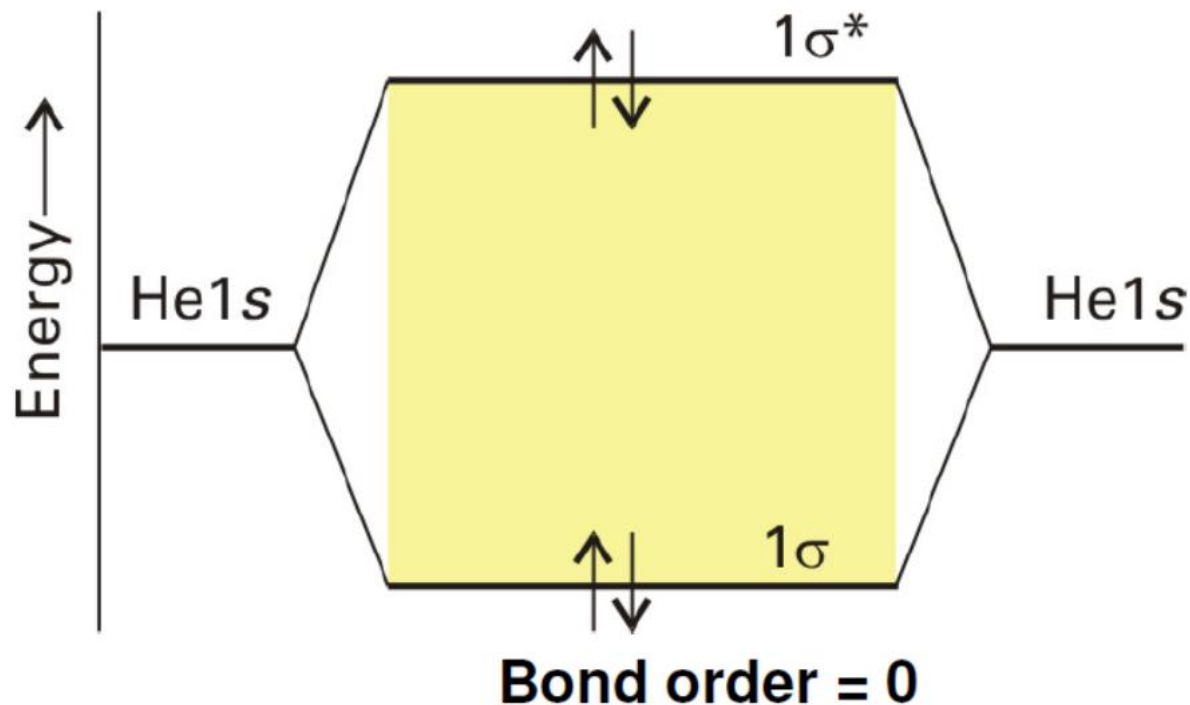
Anti-bonding orbitals – atomic orbitals overlap destructively; a nodal plane bisecting the internuclear axis; energy greater than that of the constituent atoms. e.g. σ_u , π_g , δ_u , ..., orbitals. (Also denoted by σ^* , π^* , δ^* , ...)

Non-bonding orbitals – atomic orbitals that cannot mix because of symmetric/energetic incompatibilities remain almost unaltered and are called non-bonding orbitals.

$$\text{Bond order (BO)} = (\text{N}_{\text{bonding}} - \text{N}_{\text{anti-bonding}})/2;$$

($\text{N}_{\text{bonding}}$ = no of electrons in bonding orbitals; $\text{N}_{\text{anti-bonding}}$ = no of electrons in antibonding orbitals)

He₂ molecule?



Calculate the bond order for the following molecules:

H₂, H₂⁺, H₂⁻, He₂⁺, He₂⁻. (Homework)

He₂ $1\sigma_g^2 1\sigma_u^2$ does not exist. Anti bonding orbital occupation is more destabilizing than the stabilizing effect of bonding orbital occupation.

He₂⁺ Exists. (spectroscopic evidence) Why?

Period 2 homonuclear diatomics

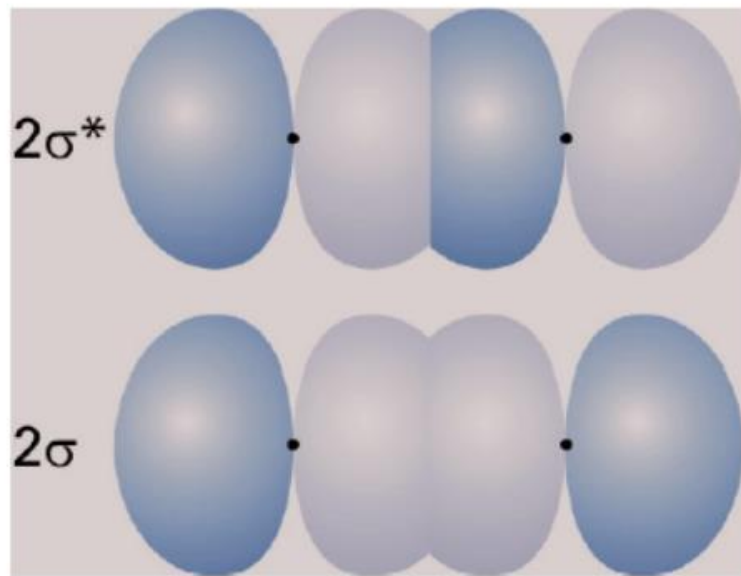


By convention, the inter-nuclear axis lies on z-axis.

The K shell (core) orbitals do not participate in bonding – these orbitals are non-bonding in nature

1σ and $1\sigma^*$ from the 2s orbitals

2σ and $2\sigma^*$ from the $2p_z$ orbitals



The bonding σ orbitals are symmetric wrt inversion, hence are represented as σ_g .

The antibonding σ orbitals (the σ^* orbitals) are antisymmetric wrt inversion, hence represented as σ_u .

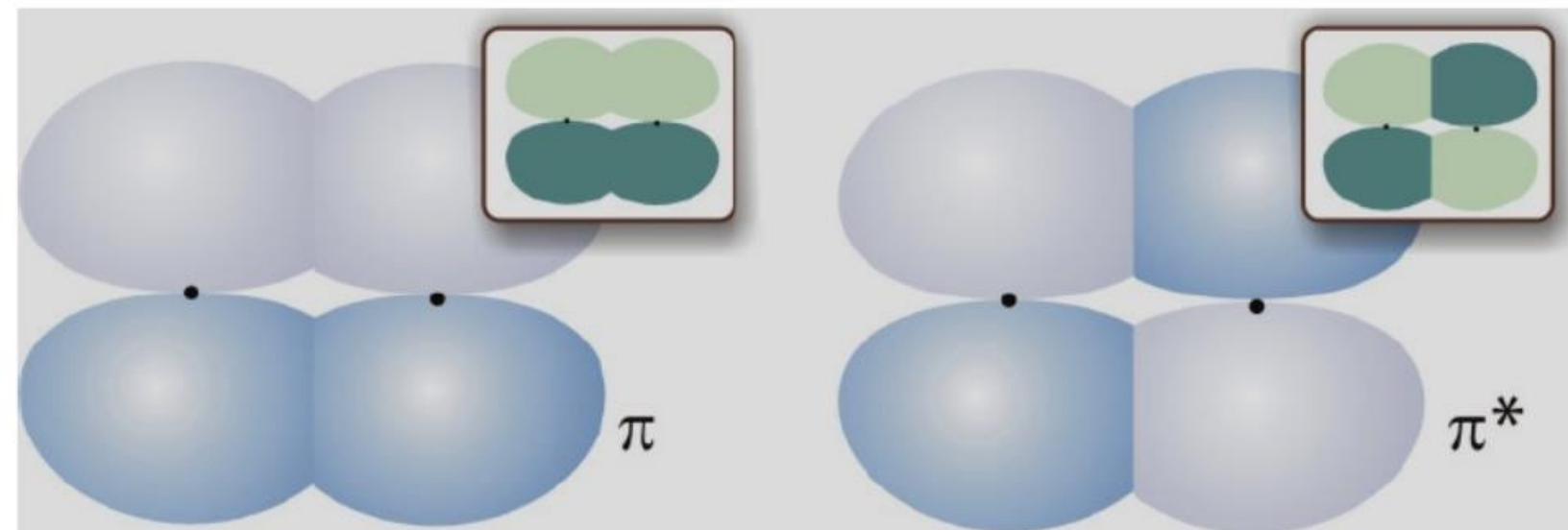
Period 2 homonuclear diatomics

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lead

The linear combinations of the two $2p_x$ (or the two $2p_y$) orbitals result in formation of π and π^* orbitals.



The bonding π orbitals are antisymmetric wrt inversion, hence are represented as π_u .

The antibonding π orbitals (the π^* orbitals) are symmetric wrt inversion, hence represented as π_g .

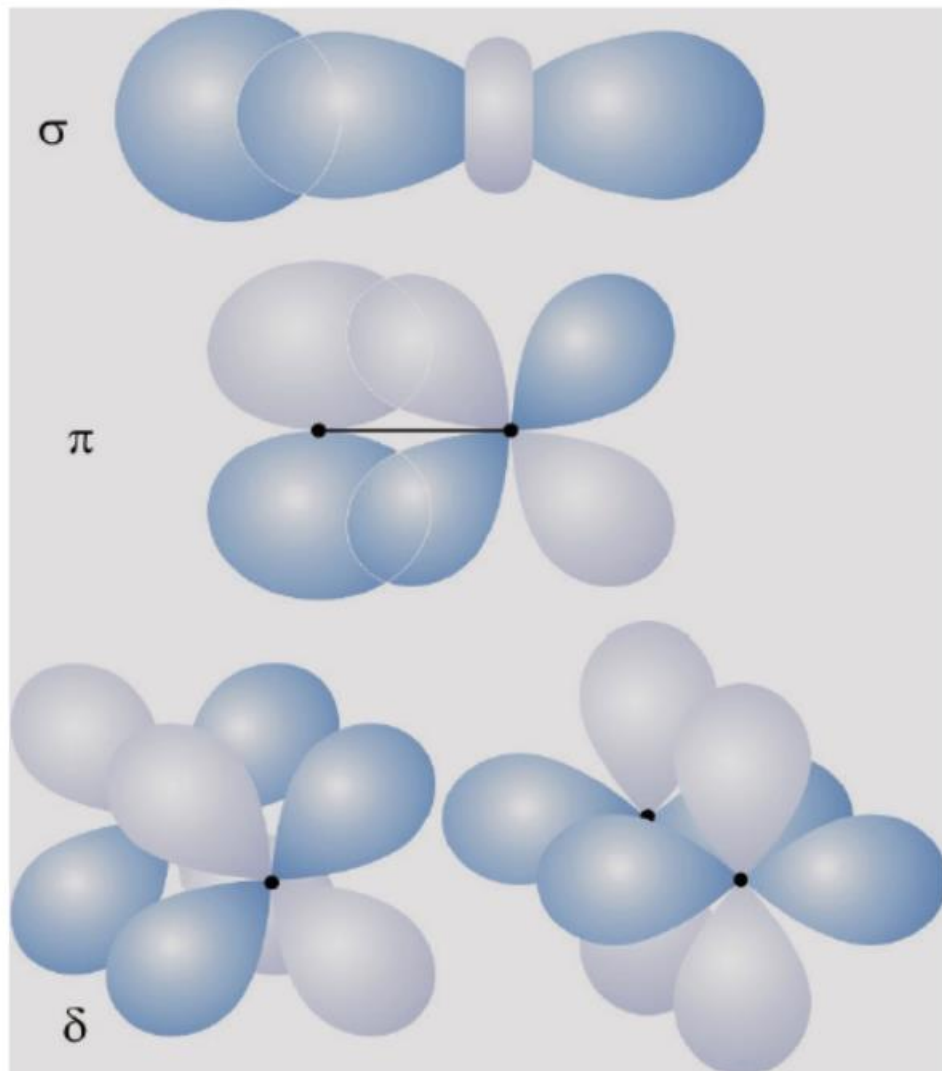
The σ -, π - and δ - orbitals

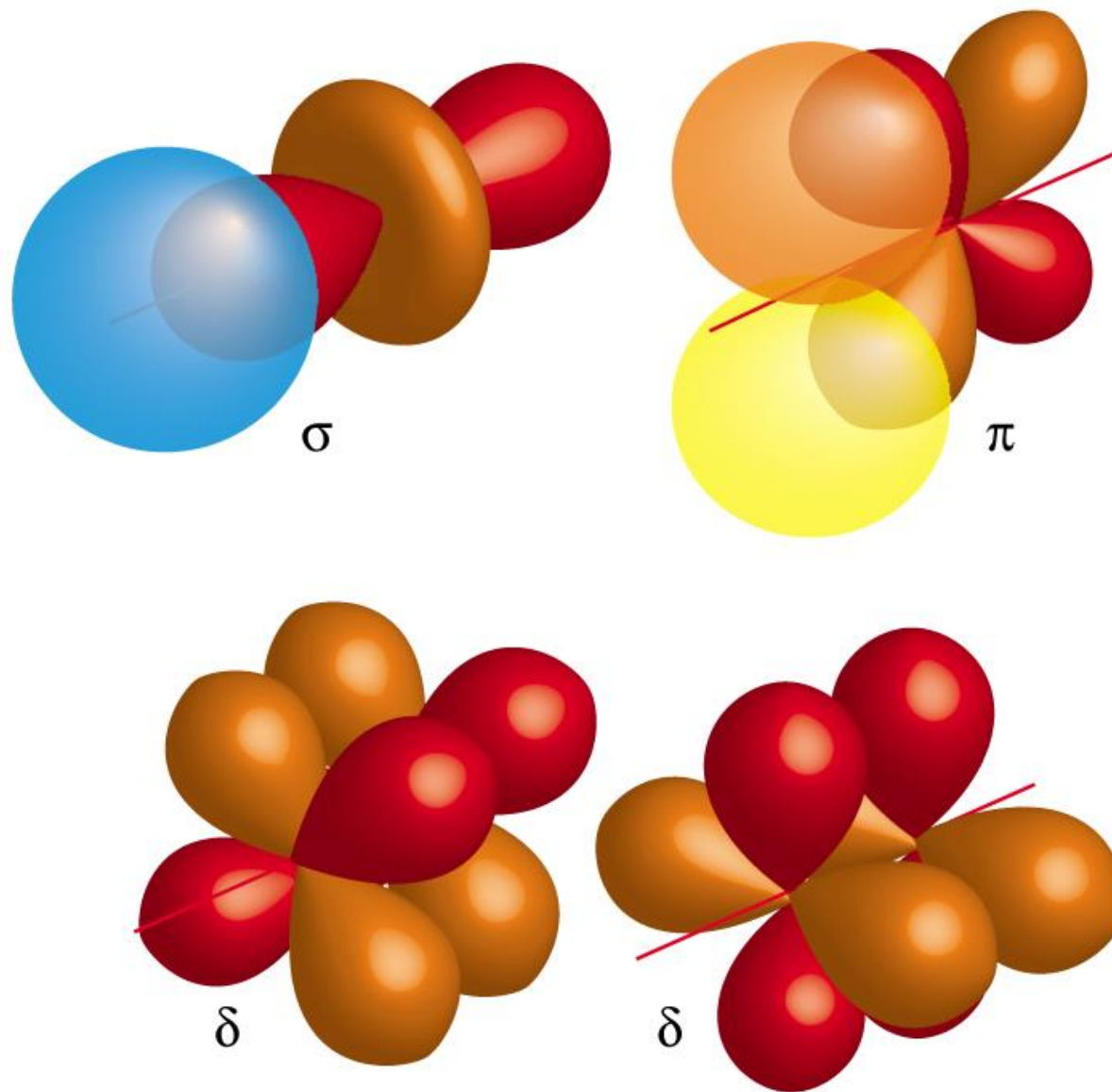
In general, diatomic molecules can have π - and δ -orbitals in addition to the σ -orbitals.

In a σ -orbital, no nodal plane contains the inter-nuclear axis.

In a π -orbital, there is one nodal plane containing the inter-nuclear axis.

In a δ -orbital, two nodal planes contain the inter-nuclear axis.





How to identify the orbitals?

Consider rotations about internuclear axis.

360° rotation and no change → σ orbital

180° rotation and the sign changes → π orbital

90° rotation and the sign changes → δ orbital.

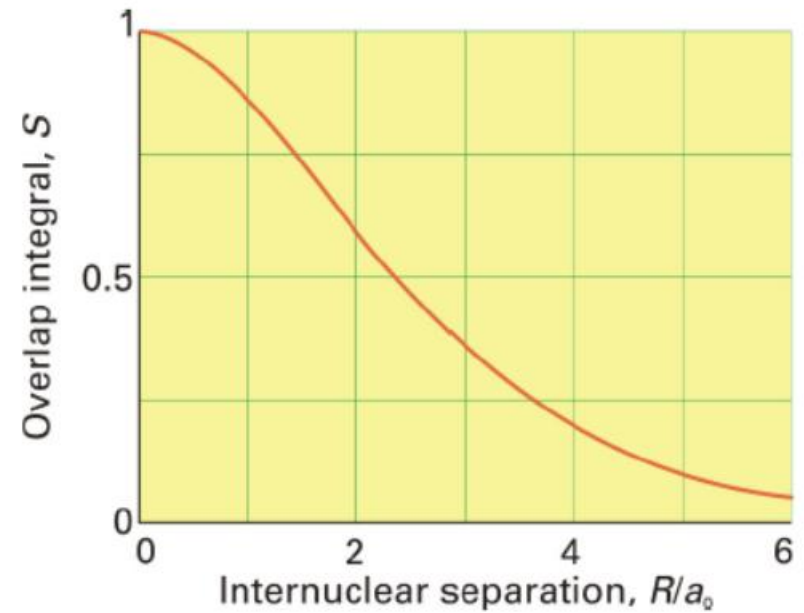
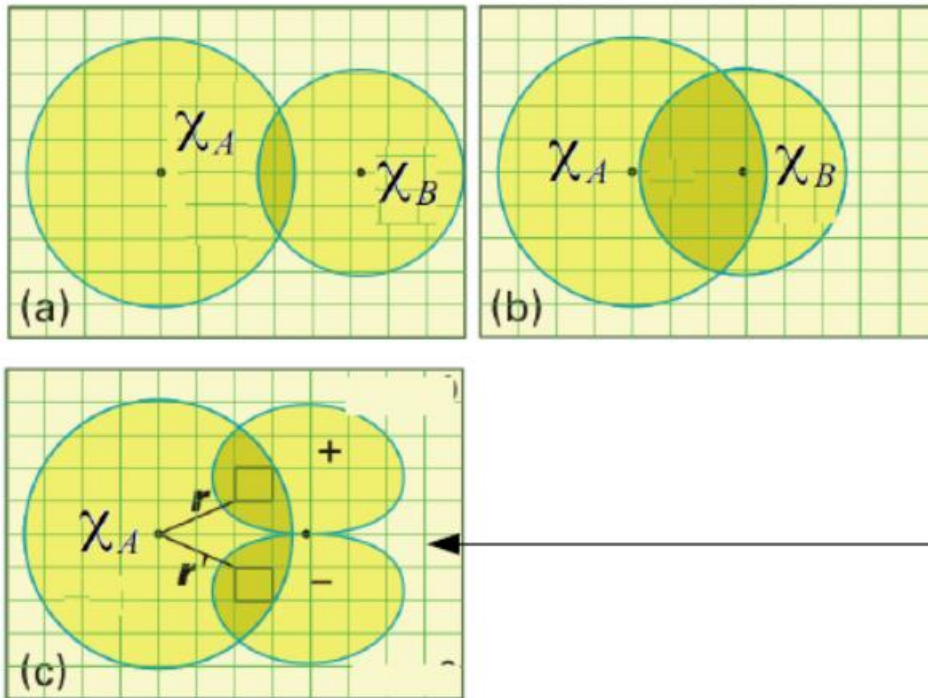
Symmetry and overlap

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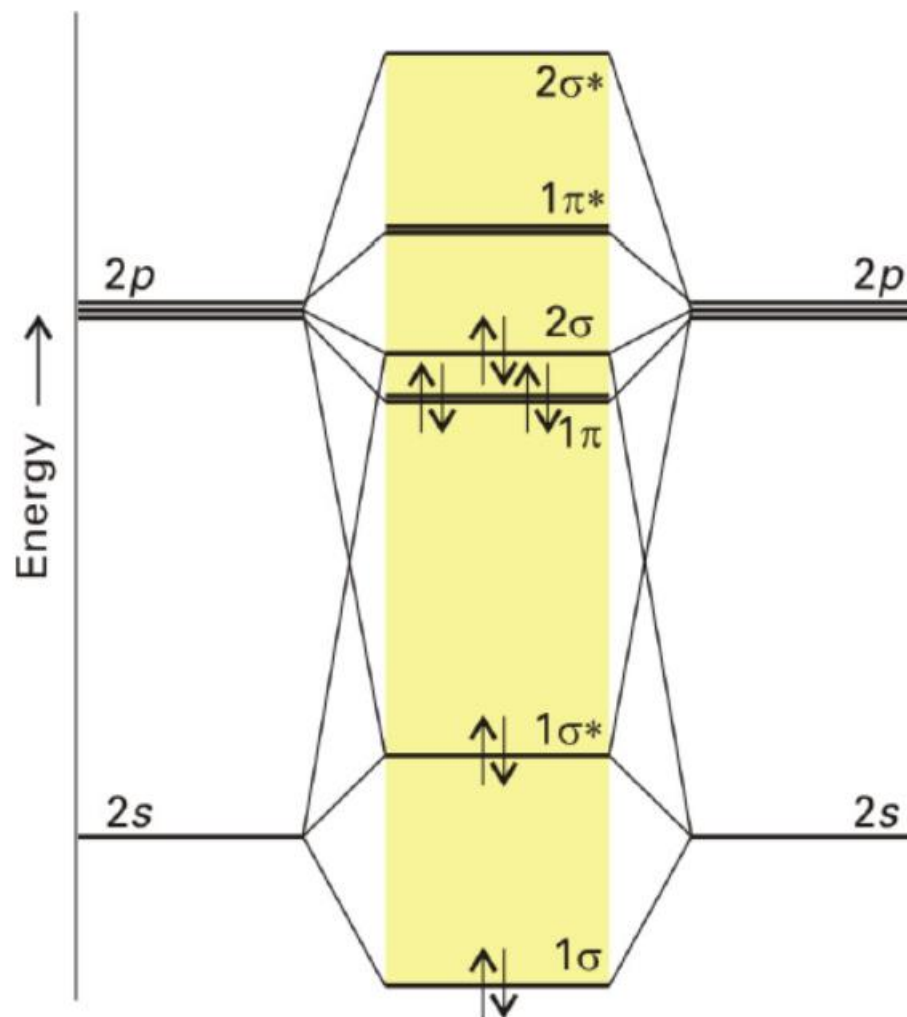
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Overlap decreases with increase in internuclear distance

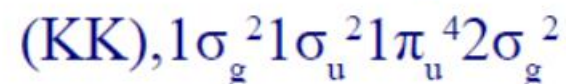


Incompatible
symmetry
arrangement

MO energy level diagram for N_2 and smaller diatomics



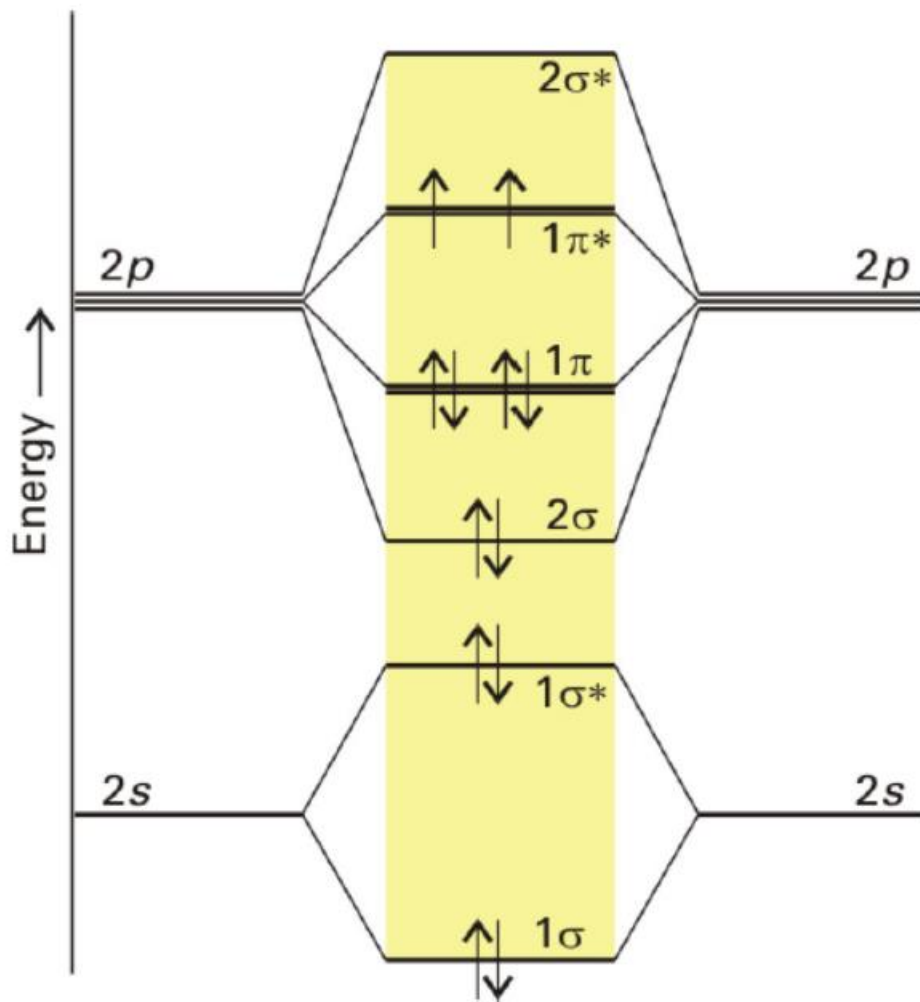
Electronic configuration:



$$\text{Bond order} = (8-2)/2=3$$

What would be the bond orders of N_2^+ and N_2^- molecules?

MO energy level diagram for O₂ and larger diatomics



Electronic configuration:

(KK), $1\sigma_g^2 1\sigma_u^2 2\sigma_g^2 1\pi_u^4 1\pi_g^2$

Bond order = $(8-4)/2=2$

What would be the bond orders of O₂⁺ and O₂⁻ molecules?

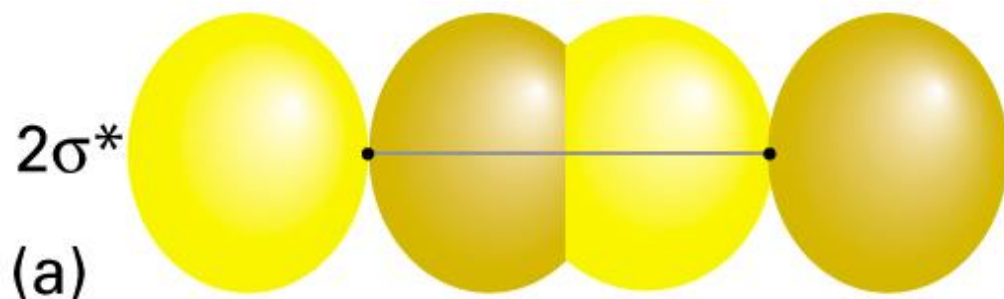
Tutorial discussion: Electronic configurations and bond-orders of second row homonuclear diatomics

Molecular Orbital Theory

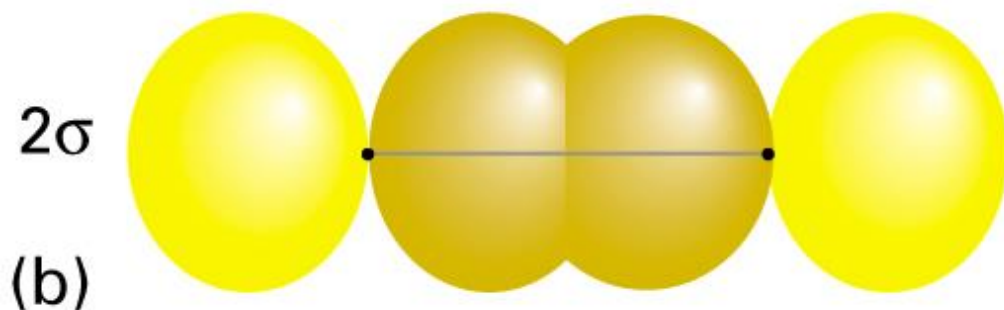


Consider O_2 molecule; Consider only valence shell;

2s orbital combinations are same as 1s orbital combinations; Consider 2p orbitals.

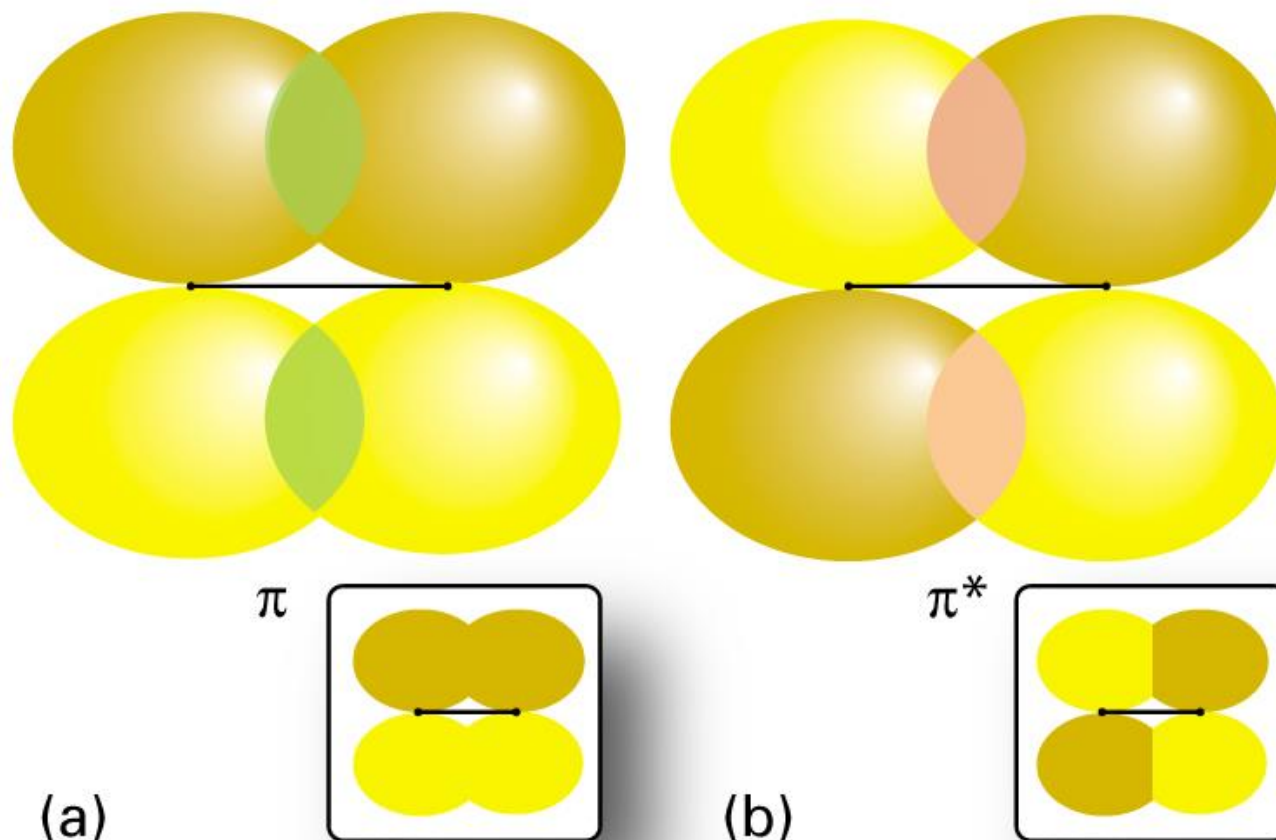


*Anti-bonding
(odd – u)*



*Bonding
(even – g)*

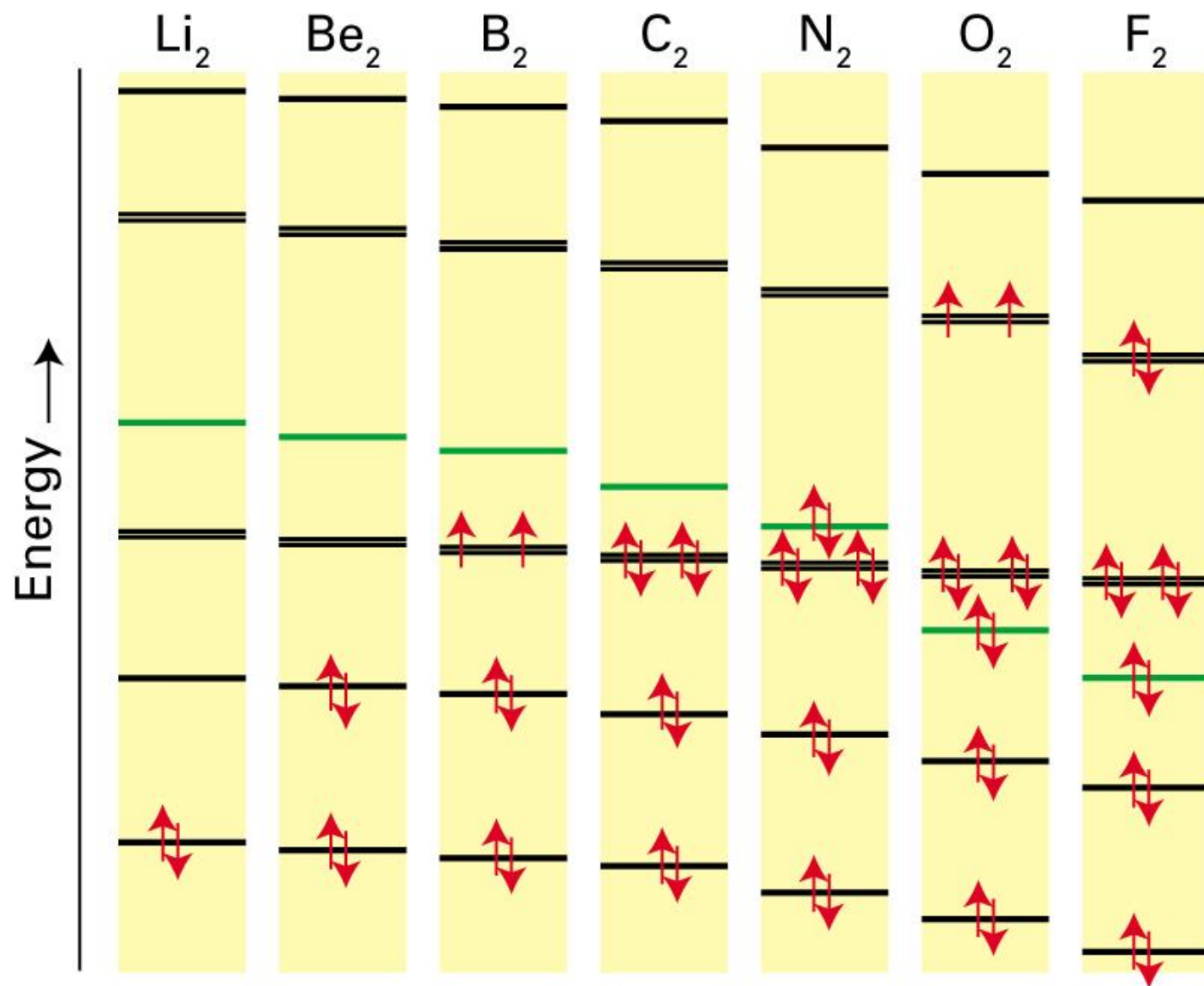
Molecular Orbital Theory



Bonding
(odd – u)

Anti-bonding
(even – g)

Period 2 homonuclear diatomics: relative MO energies



*Note: Up to N₂
the 2σ_g is above
1π_u*

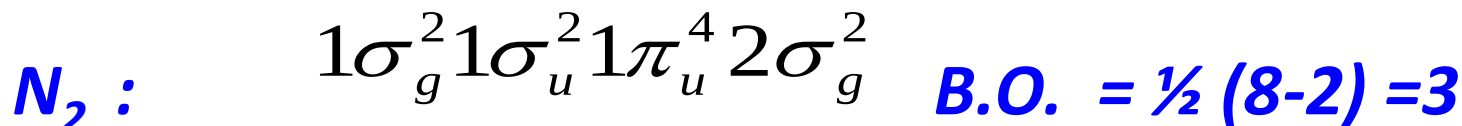
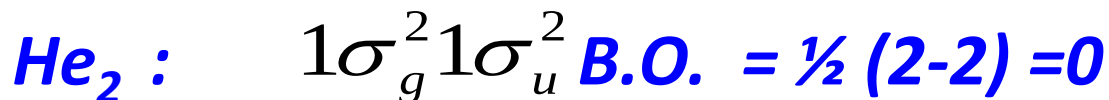
*Be₂ does not
exist; However,
Be metal exists.
(Band theory-
beyond this
course)*

Molecular Orbital Theory



Bond Order: $\frac{1}{2} (n - n^*)$

n = No. Of Bonding electrons; n^ = No. of antibonding electrons.*



Bond order is an indicative number; not energy. So while comparing be careful. Compare only that of similar species such as O_2 and O_2^+ , but not O_2^{2+} and N_2 .

Self-study

Paramagnetism and diamagnetism:

Paramagnetic substances: Attracted towards magnetic field. (unquenched spin or orbital mag. Moment)

Diamagnetic substances: Repelled by a magnetic field.

Consider Oxygen molecule : $1\sigma_g^2 1\sigma_u^2 2\sigma_g^2 1\pi_u^4 \boxed{1\pi_g^2}$

2 unpaired electrons; paramagnetic due unquenched spin magnetic moment in π^* orbitals. Success of MO Theory